Source: https://cam.feedzai.com/ds-studio-ds-api/25.0/reference/exploration/feature_stats/

feature_stats Â¶

collect_value_counts Â¶

```
collect_value_counts(data: _TDataFrame, string_fields: list[str], dist_sampling: float | None) -
> dict[str, dict[str, int]]
```

Compute sampled value counts for string fields via groupBy.

It identifies the frequency of unique values within specified string columns.

Parameters:

Name	Type	Description	Default
`data` Â¶	`DataFrame`	The source DataFrame containing string columns to analyze.	*required*
`string_fields` Â¶	`list of str`	The names of the columns for which to compute value distributions.	*required*
`dist_sampling` Â¶	`float`	The sampling fraction (0.0 to 1.0). If None or 0.0, the function returns an empty dictionary immediately.	*required*

Returns:

Type	Description
`dict of (str, dict of (str, int))`	A mapping where each key is a field name, and the value is a dictionary containing unique strings and their respective occurrence counts.

datetime_stat_exprs Â¶

```
datetime_stat_exprs(field_name: str) -> list[tuple[str, str, Column]]
```

Generate Spark aggregate expressions that compute statistics for a given datetime column.

Parameters:

Name	Type	Description	Default
`field_name` Â¶	`str`	The name of the datetime column to analyze. The name is automatically escaped using backticks for SQL safety.	*required*

Returns:

Type	Description
`list of tuple of (str, str, pyspark.sql.Column)`	A list of triples containing (field_name, stat_description, spark_expression). The statistics included are: - count: Total number of non-null datetime entries. - missing: Count of NULL values in the column. - min: The earliest (minimum) timestamp in the dataset. - max: The latest (maximum) timestamp in the dataset.

get_expected_stats_set Â¶

```
get_expected_stats_set(field_type)
```

Returns the set of stats retrieved by compute_stats for a given field type.

Parameters:

Name	Type	Description	Default
`field_type` Â¶		type of field.	*required*

Returns:

Type	Description
`unknown`	set of expected computed stats.

numerical_stat_exprs Â¶

```
numerical_stat_exprs(field_name: str) -> list[tuple[str, str, Column]]
```

Generate Spark aggregate expressions that compute statistics for a given numerical column.

Parameters:

Name	Type	Description	Default
`field_name` Â¶	`str`	The name of the numerical column to analyze. The name is automatically escaped using backticks for SQL safety.	*required*

Returns:

Type	Description
`list of tuple of (str, str, pyspark.sql.Column)`	A list of triples containing (field_name, stat_description, spark_expression). The statistics included are: - sum: Sum of all finite values. - min: Minimum finite value. - max: Maximum finite value. - nan: Count of NaN values. - zeros: Count of values equal to 0. - infinites: Count of positive and negative infinity values. - missing: Count of null values. - count: Count of finite values. - mean: Arithmetic mean of finite values. - var: Sample variance of finite values.

string_stat_exprs Â¶

```
string_stat_exprs(field_name: str) -> list[tuple[str, str, Column]]
```

Generate Spark aggregate expressions that compute statistics for a given string column.

Parameters:

Name	Type	Description	Default
`field_name` Â¶	`str`	The name of the string column to analyze. The name is automatically escaped using backticks for SQL safety.	*required*

Returns:

Type	Description
`list of tuple of (str, str, pyspark.sql.Column)`	A list of triples containing (field_name, stat_description, spark_expression). The statistics included are: - count: Count of values that are neither NULL nor empty strings. - empty: Count of empty string ("") values. - missing: Count of NULL values. - hll_cardinality: An approximate count of distinct values using the HyperLogLog++ algorithm for memory efficiency.

validation_exprs Â¶

```
validation_exprs(field_name: str, validations: list[Validation]) -> list[tuple[str, str, Column]]
```

Build Spark aggregate expressions for custom validations on a field..

Iterates over a provided set of validations and generates a SQL expression that encompasses all provided validations.

Parameters:

Name	Type	Description	Default
`field_name` Â¶	`str`	The name of the column to be validated. The name is automatically escaped using backticks for SQL safety.	*required*
`validations` Â¶	`list of Validation`	A list of Validation objects, each containing a description and a 'sql_builder' method that accepts a Column and returns a Spark aggregate expression.	*required*

Returns:

Type	Description
`list of tuple of (str, str, pyspark.sql.Column)`	A list of triples containing (field_name, validation_description, spark_expression). These are typically collected into a single data.agg() call alongside standard statistics.